

Comparative Analysis of Machine Learning Algorithms for Breast Cancer Diagnosis

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Abstract—This paper presents a comparative analysis of supervised machine learning algorithms for breast cancer diagnosis using the Wisconsin Breast Cancer dataset. The objective of this study is to evaluate and compare the performance of different classification models and to analyze the impact of data Pre-processing techniques on their effectiveness.

Four classification algorithms were used: logistic regression, KNN (k-Nearest Neighbors) algorithm, random forest and eXtreme Gradient Boosting (XGBoost). The dataset was pre-processed by handling missing values, standardizing features, handling outliers and preparing the data for model training.

The performance of the applied models was evaluated using accuracy as the primary metric. After training, we found that the logistic regression model achieved the highest accuracy of 0.974, outperforming all other models. Both the K-Nearest Neighbors model and the Random Forest model showed similar performance, with an accuracy of approximately 0.965. On the other hand, the XGBoost model achieved the lowest accuracy of 0.956.

These results indicate that the dataset exhibits relatively simple patterns that can be effectively detected using linear models. However, more complex models do not necessarily perform better in this case. This study shows the importance of selecting right machine learning models and Pre-processing strategies for medical diagnosis tasks and provides a clear comparison of algorithm performance.

Index Terms—Breast Cancer, Machine Learning, Classification, Logistic Regression, K-Nearest Neighbors, Random Forest, eXtreme Gradient Boosting

I. INTRODUCTION

Breast cancer is one of the most common and dangerous diseases of our time, so early detection is important for improving patient survival rates. Progress in medical data collection and analysis has enabled the use of machine learning techniques to support disease diagnosis and decision-making processes.

Machine learning models are able to find complex patterns on medical data and can give accurate predictions. However, their performance depends on several factors, including data preprocessing, feature scaling, and the choice of algorithms. Different models can produce different results on the same dataset, demonstrating the necessity of comparative analysis.

In this study, we focus on the classification of breast cancer using the Wisconsin Breast Cancer dataset. This dataset contains several diagnostic features extracted from digitized images of breast cell nuclei, which are used to determine whether a tumor is benign or malignant.

The main objective of this work is to compare the performance of three used classification algorithms: Logistic Regression, K-Nearest Neighbors (KNN), Random Forest, and eXtreme Gradient Boosting (XGBoost). Additionally, this study investigates the impact of processing techniques, particularly feature scaling, on model performance.

The results of this study aim to provide a better understanding of the strengths and weaknesses of each algorithm and to identify the most suitable model for the detection of breast cancer.

II. STATE OF THE ART

Machine learning may be one of the most promising fields for future advancements in medicine, improving diagnosis and aiding decision-making processes. We applied a multi-machine learning algorithm to breast cancer data. As we all know, breast cancer detection has received considerable attention due to its impact on public health, so we used the machine learning algorithm to classify tumors as benign or malignant based on clinical characteristics.

One of the algorithms we used is logistic regression, a basic model for binary classification problems. It may be good in terms of simplicity, interpretability, and efficiency, but it has weaknesses in that it cannot capture complex nonlinear relationships in the data.

We also used the K-Nearest Neighbors (KNN) algorithm, another common classification algorithm based on distance metrics, which ranks data points according to their nearest neighbors, making them a sample to be used. However, on the other hand, the K-Nearest Neighbors (KNN) algorithm is sensitive to feature scaling and the selection of the number of neighbors (K), and its performance can decrease with large datasets.

There is also the random forest algorithm, a collective learning technique that combines several decision trees to improve predictive performance. This model is known for its robustness, ability to handle complex data, and resistance to overfitting. However, it is less interpretable compared to simpler models such as logistic regression.

Finally, we used XGBoost, an advanced clustering technique based on gradient enhancement. This technique is known for its high performance and efficiency in handling structured

data. Despite its advantages, XGBoost requires careful hyperparameters and may not always outperform simpler models, depending on the dataset.

Previous studies in medical diagnosis have shown that no single algorithm consistently outperforms others across all datasets. Therefore, comparative analysis of multiple models is essential to determine the most suitable approach. This motivates the present study, which evaluates and compares the performance of Logistic Regression, KNN, Random Forest, and XGBoost for breast cancer classification.

III. METHODOLOGY

This section describes the dataset used, the Pre-processing steps applied, and the machine learning models chosen for the classification task.

A. Dataset

We trained our models using the Wisconsin Breast Cancer Dataset. This dataset contains numerous digital properties extracted from digital images of breast cell nuclei. These properties describe attributes such as radius, texture, circumference, area, and smoothness.

The objective of the dataset is to classify tumors into two categories: benign (non-cancerous) and malignant (cancerous). The dataset is widely used in machine learning research for evaluating classification models in medical diagnosis.

B. Data Pre-processing

Pre-processing the data had a significant impact on improving model performance. We did several steps to prepare the dataset:

- **Data Cleaning:** Non-helpful attributes such as identifiers were removed from the dataset.
- **Encoding:** The target variable was transformed into numerical values, where malignant cases were encoded as 1 and benign cases as 0.
- **Handling Missing Values:** The dataset was checked for missing values and none were found.
- **Outlier Detection:** Outliers were identified using the Interquartile Range (IQR) method. Although several features contained outliers, they were retained as they may represent important variations in medical data.
- **Feature Scaling:** Standardization was applied using the StandardScaler to normalize the feature values, which is particularly important for distance-based algorithms such as KNN.
- **Train-Test Split:** The dataset was divided into training and testing sets using an 80/20 split to evaluate model performance.

C. Model Selection

Four classification algorithms were selected for this study:

- **Logistic Regression:** Used as a baseline model due to its simplicity and effectiveness in binary classification.
- **K-Nearest Neighbors (KNN):** A distance-based algorithm where the optimal number of neighbors ($K = 9$) was determined using the elbow method.

- **Random Forest:** An ensemble method that combines multiple decision trees to improve prediction accuracy and reduce overfitting. Hyperparameter tuning was performed using Randomized Search, resulting in optimal values of `n_estimators=150`, `max_depth=10`, and `min_samples_split=5`. The parameter `random_state=42` was kept fixed to ensure reproducibility.
- **XGBoost:** An advanced gradient boosting algorithm known for its high performance. Hyperparameter tuning using Randomized Search yielded optimal values of `n_estimators=150`, `max_depth=3`, and `learning_rate=0.1`. The parameter `random_state=42` was maintained to ensure reproducibility of results.

D. Workflow Description

Our methodology follows these steps: data loading, pre-processing, model training, prediction, and evaluation. Each model was trained on the training set and evaluated on the test set using multiple performance metrics.

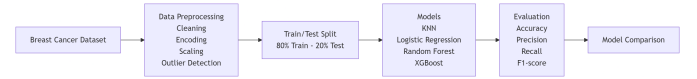


Fig. 1. Machine Learning Pipeline for Breast Cancer Classification

IV. APPLICATION

This section describes the implementation of the proposed approach and the tools used during the experiments.

A. Implementation

Python programming language was used to implement the project. The procedure has many processes such as data loading, preprocessing, training, and evaluation. The implementation of the proposed approach was carried out using Python. The complete source code and experiments are publicly available in a GitHub repository [1], ensuring transparency and reproducibility of the results. At first, the dataset was loaded and preprocessed by eliminating unnecessary columns and transforming categorical labels into numerical values. Techniques like scaling of data and identifying outliers were used during the pre-processing phase.

The uploaded dataset was divided into two sets: a training set and a test set, with 80

In the case of the K-nearest neighbor classifier, the number of neighbors was optimized by applying the elbow technique. All the implemented algorithms were trained using the training set and evaluated using performance indicators such as accuracy, precision, recall, and F1 score.

Finally, comparisons of all algorithms were done to choose the most efficient strategy for classifying breast cancer.

B. Tools Used

The experiments were conducted using the Python programming language along with several libraries:

- **Pandas:** Used for data manipulation and preprocessing.
- **NumPy:** Used for numerical computations.
- **Scikit-learn:** Used for implementing machine learning models such as Logistic Regression, KNN, and Random Forest.
- **XGBoost:** Used for implementing the gradient boosting model.
- **Matplotlib:** Used for data visualization, including performance graphs and model comparison.

V. RESULTS AND DISCUSSION

A. Model Performance Comparison

Table I presents the performance of the implemented models using accuracy as the evaluation metric.

TABLE I
MODEL PERFORMANCE COMPARISON

Model	Accuracy
Logistic Regression	0.974
KNN	0.965
Random Forest	0.965
XGBoost	0.956

B. Graphical Analysis

Figure 2 shows the accuracy of each model using a graphical representation.

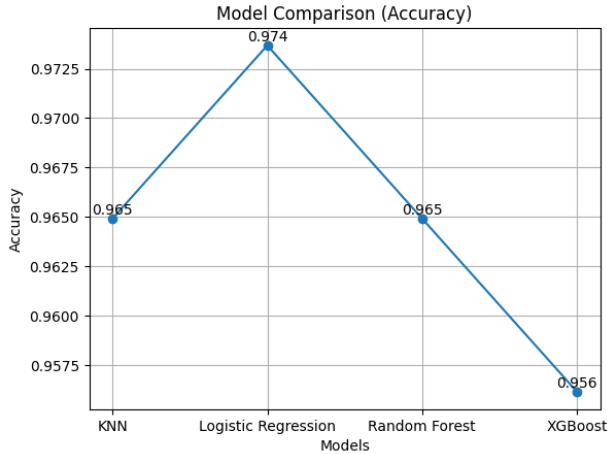


Fig. 2. Accuracy comparison of machine learning models

C. Discussion

As we can see from the results, logistic regression achieved the highest accuracy of 0.974, making it the best model for this type of data. This means that the dataset exhibits patterns that can be effectively captured using a linear model.

K-Nearest Neighbors (KNN) algorithm and the Random Forest achieved the same performance, with an accuracy of

0.965. While these models are able to capture more complex relationships, they did not improve the results compared to logistic regression.

On the other hand, XGBoost achieved a low accuracy of 0.956. Although hyperparameter tuning was applied, its performance remained lower than the other models. This may indicate that the dataset does not require highly complex models or that further tuning could be explored.

Based on these results, we can conclude that more complex models do not necessarily lead to better result. In this case, we found that the logistic regression model, a simpler model, proved more effective. This highlights the importance of selecting the optimal model and the appropriate dataset in machine learning.

VI. CONCLUSION

This paper presented a comparative analysis of multiple machine learning algorithms for breast cancer classification using the Wisconsin dataset. The main objective was to evaluate the performance of different models and analyze the impact of pre-processing techniques.

The study results showed that logistic regression achieved the highest accuracy, making it the most effective for this type of dataset, outperforming more complex models such as KNN, random forest, and XGBoost. This means that the dataset can be modeled efficiently using simpler linear methods.

The KNN and Random Forest algorithms demonstrated competitive performance, while XGBoost showed lower accuracy, possibly due to the lack of appropriate hyperparameters. These results highlight that model complexity does not always guarantee better performance, and that selecting the right model is essential for the right data.

A. Perspectives

Future work may focus on improving model performance through advanced techniques such as hyperparameter tuning, cross-validation, and feature selection. Additionally, other algorithms such as neural networks could be explored to further enhance classification accuracy.

VII. REFERENCES

REFERENCES

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